

LAB_07_Shiguihara

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1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library(DBI)
library(RSQLite)

conn <- dbConnect(SQLite(), "database.sqlite3")
```

1.1 conferindo a database

```
tabelas <- dbListTables(conn)
print(tabelas)
```

```
[1] "course_offerings"      "courses"               "grade_distributions"
[4] "instructors"           "rooms"                  "schedules"
[7] "sections"              "subject_memberships"   "subjects"
[10] "teachings"
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

```
SELECT i.name AS NOMES_PROFESSORES, COUNT(DISTINCT i.name) AS QUANTIDADE
FROM subjects s
INNER JOIN subject_memberships m ON s.code = m.subject_code
INNER JOIN course_offerings c ON m.course_offering_uuid = c.uuid
INNER JOIN sections se ON c.uuid = se.course_offering_uuid
INNER JOIN teachings t ON se.uuid = t.section_uuid
INNER JOIN instructors i ON t.instructor_id = i.id
WHERE s.abbreviation = 'STAT'
GROUP BY i.name
UNION ALL
```

```

SELECT 'QUANTIDADE TOTAL' , COUNT(DISTINCT i.id)
FROM subjects s
INNER JOIN subject_memberships m ON s.code = m.subject_code
INNER JOIN course_offerings c ON m.course_offering_uuid = c.uuid
INNER JOIN sections se ON c.uuid = se.course_offering_uuid
INNER JOIN teachings t ON se.uuid = t.section_uuid
INNER JOIN instructors i ON t.instructor_id = i.id
WHERE s.abbreviation = 'STAT';

```

Table 1: Displaying records 1 - 10

NOMES_PROFESSORES	QUANTIDADE
ABIGAIL BENZINE	1
ADIN-CRISTIAN ANDREI	1
AKICHIKA OZEKI	1
ALAN HUANG	1
ALBERTO DEL PIA	1
ALBRECHT KLEMM	1
ALEXANDER COVINGTON	1
ALEXANDER FISH	1
ALEXANDER KISELEV	1
ALYSSA DIGILIO	1

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

- Quem é o professor mais difícil?

```

SELECT i.name AS NOME_PROFESSOR, SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0)
AS MEDIA
FROM subjects s
INNER JOIN subject_memberships m ON s.code = m.subject_code
INNER JOIN course_offerings c ON m.course_offering_uuid = c.uuid
INNER JOIN grade_distributions gd ON c.uuid = gd.course_offering_uuid
INNER JOIN sections se ON c.uuid = se.course_offering_uuid
INNER JOIN teachings t ON se.uuid = t.section_uuid
INNER JOIN instructors i ON t.instructor_id = i.id
WHERE s.abbreviation = 'STAT'
GROUP BY i.name

```

```
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + g
ORDER BY MEDIA asc
LIMIT 1
```

Table 2: 1 records

NOME_PROFESSOR	MEDIA
JAMES D KUELBS	2.607692

- Quem é o professor mais fácil?

```
SELECT i.name AS NOME_PROFESSOR, SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + g
AS MEDIA
FROM subjects s
INNER JOIN subject_memberships m ON s.code = m.subject_code
INNER JOIN course_offerings c ON m.course_offering_uuid = c.uuid
INNER JOIN grade_distributions gd ON c.uuid = gd.course_offering_uuid
INNER JOIN sections se ON c.uuid = se.course_offering_uuid
INNER JOIN teachings t ON se.uuid = t.section_uuid
INNER JOIN instructors i ON t.instructor_id = i.id
WHERE s.abbreviation = 'STAT'
GROUP BY i.name
ORDER BY MEDIA desc
LIMIT 3
```

Table 3: 3 records

NOME_PROFESSOR	MEDIA
YAJUAN SI	4
SRIKANTHMADHAVAN ARAVAMUTHAN	4
GUANHUA CHEN	4

- Qual é a disciplina mais difícil?

```
SELECT c.name AS DISCIPLINA , SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + g
AS MEDIA
FROM subjects s
INNER JOIN subject_memberships m ON s.code = m.subject_code
INNER JOIN course_offerings c ON m.course_offering_uuid = c.uuid
INNER JOIN grade_distributions gd ON c.uuid = gd.course_offering_uuid
```

```

INNER JOIN sections se ON c.uuid = se.course_offering_uuid
INNER JOIN teachings t ON se.uuid = t.section_uuid
INNER JOIN instructors i ON t.instructor_id = i.id
WHERE s.abbreviation = 'STAT'
GROUP BY DISCIPLINA
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + g
ORDER BY MEDIA asc
LIMIT 1

```

Table 4: 1 records

DISCIPLINA	MEDIA
Intro-Theory of Probability	2.896207

- Qual é a disciplina mais fácil?

```

SELECT c.name AS DISCIPLINA , SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + g
AS MEDIA
FROM subjects s
INNER JOIN subject_memberships m ON s.code = m.subject_code
INNER JOIN course_offerings c ON m.course_offering_uuid = c.uuid
INNER JOIN grade_distributions gd ON c.uuid = gd.course_offering_uuid
INNER JOIN sections se ON c.uuid = se.course_offering_uuid
INNER JOIN teachings t ON se.uuid = t.section_uuid
INNER JOIN instructors i ON t.instructor_id = i.id
WHERE s.abbreviation = 'STAT'
GROUP BY DISCIPLINA
ORDER BY MEDIA DESC
LIMIT 4

```

Table 5: 4 records

DISCIPLINA	MEDIA
Sample Survey Theory&Method	4
Nonparam Stats&Machine Learng	4
Empir Proc&Semiparmtrc Infernc	4
Data Science Practicum	4

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

Horário de Compilação

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-23 15:13:57 -03"
```